

Ibn Tofail University

Rigid Body Mechanics

Problem Set II

Course Question*

Show that the velocity vector field of a solid defines a torsor. Specify its reduction elements.

Correction

Exercise 1:

In the following solid motions, define the Euler angles, deduce the number of degrees of freedom, and determine the corresponding instantaneous rotation velocities.

1) A rod OA with negligible cross-section rotating around the axis $(O, \vec{z}_o)$ in the plane $(O, \vec{x}_o, \vec{y}_o)$.

2) A disk of center D linked to a rod OA (D coincides with A and the rod OA is contained in the plane of the disk). The motion takes place in the plane $(O, \vec{x}_o, \vec{y}_o)$.

a) The connection between the disk and the rod is rigid.

b) The connection between the disk and the rod allows the disk to perform a rotation around the axis $(A, \vec{z})$ (fig.1).

3) A solid (S) consisting of a hemisphere of radius a and base center O_2 rigidly attached to a rod OO_2 (fig.2). The solid is mobile around point O.

4) A homogeneous ball (B), of mass m, of radius R, of center G, moving on a fixed horizontal plane (O, X, Y) .

Correction

Exercise 2:

Consider a robot consisting of a base S_0 and two arms 1 and 2 (fig.3).

Let the reference frames be:

$R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$ fixed reference frame linked to the base S_0 .

$R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_1)$ reference frame linked to arm 1.

$R_2(A, \vec{x}_2, \vec{y}_2, \vec{z}_2)$ reference frame linked to arm 2.

Given:

$$\overrightarrow{OA} = \ell_1 \vec{x}_1 \quad \text{and} \quad \overrightarrow{AB} = \ell_2 \vec{x}_2$$

- 1) Calculate the instantaneous rotation velocities $\vec{\Omega}(R_1/R_o)$ and $\vec{\Omega}(R_2/R_o)$.
- 2) Calculate the velocity $\vec{V}(B/R_o)$ and the acceleration $\vec{\gamma}(B/R_o)$.

Correction

Exercise 3:

A rectangular plate OABC is mobile with respect to a reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$ such that its vertex O remains fixed in R_o . The edge OA remains in the plane $(O, \vec{x}_o, \vec{y}_o)$ with $OA = a$ and $AB = b$.

- 1) Determine the parameters defining the motion of the plate.
- 2) Find the velocity field of the points of the plate (case of points A, B, C).

Correction

Exercise 4:

Consider (fig.4) the system (S) formed by a horizontal rod AB of length a , and a vertical circular disk perpendicular to AB, of center B, of radius R , in motion in a fixed reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$.

Point A is fixed on the axis $(O, \vec{z}_o)$ at distance R from O. Let I be the point of the disk in contact with the horizontal plane $(O, \vec{x}_o, \vec{y}_o)$, and I^* the geometric contact point.

We call $(O, \vec{u})$ the axis determined by the vector $\overrightarrow{OI^*}$ and $(O, \vec{v})$ the axis such that the triad $(O, \vec{u}, \vec{v}, \vec{z}_o)$ is direct, and $(B, \vec{y})$ and $(B, \vec{z})$ two orthogonal axes linked to the disk and located in its plane. Let M be the point of the disk such that $\overrightarrow{BM} = R\vec{y}$. We denote $(\vec{x}_o, \vec{u}) = \theta$, measured around $\vec{z}_o$ and $(\overrightarrow{BI^*}, \vec{y}) = \phi$, measured around $\vec{u}$ (fig.5). We call R the reference frame $R(B, \vec{u}, \vec{y}, \vec{z})$ and R_1 the reference frame $(O, \vec{u}, \vec{v}, \vec{z}_o)$.

- 1) Determine the number of parameters necessary to describe the motion of the system.
- 2) Calculate the instantaneous rotation vectors $\vec{\Omega}(R_1/R_o)$, $\vec{\Omega}(R/R_1)$ and $\vec{\Omega}(R/R_o)$.
- 3) Study of the motion of R with respect to R_o : Determine $\vec{V}(M/R_o)$, $\vec{V}(I/R_o)$, $\vec{V}(I/R_1)$ and $\vec{\gamma}(I/R_o)$.
- 4) Write the condition of rolling without slipping at I and integrate it assuming that $a = 2R$, $\dot{\theta} = \omega = \text{const}$, and that for $t = 0$ we have $\theta = 0$ and $\phi = 0$. Show that then $\vec{\Omega}(R/R_o) = \lambda \overrightarrow{AI^*}$, where λ is a constant to be determined, and that the motion is tangent to a rotation around $\overrightarrow{AI^*}$.

Correction